Total Printed Pages: 02

Roll No. :

ECA 101/ EIA 101

Examination – Dec - 2024

B.Tech. I Semester EC/EI

Basic Electronics

Time : 3 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

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Q.1	(a)	i. Semiconductor acts as an insulator in the presence of impurities. (a) True (b) False ii. What are the charge carriers in semiconductors? (a) Electrons and holes (b) Electrons (c) Holes	02	1	1
	(b)	How minority charge-carrier produced in Si wafer doped with n-type impurities. Draw its characteristics.	02	2	1
	(c)	What is Zener diode ? What are its applications?	03	1	1
	(d)	What is Tunneling? Draw and explain its working and V-I characteristics.	05	3	1
		OR			
		What do you mean by rectification? Draw and explain the most practical circuit of the full-wave rectifier..	05	3	4
Q.2	(a)	i. BJT is a current controlled device (a) True (b) False ii. The working regions/modes of BJT is	02	1	2
	(b)	What is the purpose of doping the base region in a Bipolar junction transistor (BJT) at a lower level compared to the emitter region?	02	3	2
	(c)	What do you understand by α and β in relation to transistor?	03	2	4
	(d)	Explain the circuit diagram and working of a emitter follower.	05	3	2
		OR			
		Draw and discuss the input and output characteristics of a transistor.	05	3	2
Q.3	(a)	i. Which of the following pairs of h-parameters has the same dimensions: (a) h_{11} and h_{12} (b) h_{22} and h_{21} (c) h_{12} and h_{21} (d) h_{11} and h_{22} ii. h11 is a short circuit input impedance (a) True (b) False	02	1	3

	(b)	Mention the benefits of the h-parameters.	02	1	3
	(c)	The input impedance to a common-emitter transistor amplifier is 1.2kΩ with β = 120, r_o =50kΩ and load resistance is 2.7kΩ. calculate the value of r_e.	03	3	3
	(d)	Derive the h-parameters of a two port network.	05	3	3
		OR			
		Draw and discuss the r_e model of the common-emitter transistor amplifier.	05	3	3
Q.4	(a)	i. What is the purpose of RC coupling (a) To block ac (b) To block dc ii. The voltage gain expressed practically in	02	1	2
	(b)	Differentiate single stage and multistage stage amplifiers.	02	1	2
	(c)	List the different types of coupling used in multistage amplifiers.	03	1	2
	(d)	Draw and discuss the two-stage RC coupled amplifier. Mention the limitations and applications of the RC coupled amplifier.	05	2	2
		OR			
		Three amplifier stages are connected in cascade with 0.05V peak to peak input providing 150V peak-to-peak output. If the voltage gain of first stage is 20 and input to third stage is 15V peak-to-peak determine the overall voltage gain.	05	3	2
Q.5	(a)	i. ---------- type of amplifier considers certain frequencies and rejects certain frequencies. ii. Which of the following are the components of tunned amplifier? (a) Resistor (b) Capacitor (c) Inductor (d) all	02	1	2
	(b)	Draw the block diagram of a double tuned amplifier..	02	2	2
	(c)	Mention the advantages and disadvantages of the single tuned amplifiers.	03	1	2
	(d)	What do you mean by stagger and neutralization in tuned amplifier?	05	3	2
		OR			
		Discuss the transformer coupled audio power amplifier with the help of circuit diagram.	05	3	2

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